

NATIONAL RESEARCH THRUSTS IN PHYSICS – A DISCUSSION

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Physics, as you all know, is the most basic of all the natural sciences. For the past 50 years or so, its concepts, theories and methods, and instruments helped revolutionize the other natural sciences and made it possible to achieve progress in all other sciences.

Through the application of Physics and the collaboration of Physicists in various fields, Physics has directly or indirectly led to the development of such new technologies as microelectronics, lasers, computers, geodetic engineering, material technology, robotics, space technology, etc.

We can rightly and probably claim that modern science and technology are largely shaped by and automatically based on the basic science of physics. Without atomic and molecular physics, solid state physics, nuclear physics, lasers and statistical devices, the advance in modern science and technology would have been inconceivable.

It is the recognition of this fundamental revolutionary innovative character of physics that has made this discipline occupy the central position in the various technological plans and programs of the advanced countries of the world. The high impact of technological innovation and high growth of science-based industry that continually sprout progress have proven time and time again to these the advanced countries that it pays to invest in physics. It is therefore a source of frustration to us in the lower physics profession that our discipline has been neglected by the government.

It is indeed incomprehensible that in all the fields of science and technology, physics is the most neglected and the most backward discipline in our country today.

Physics in the Philippines is so backward that in my estimate, we are 15 years behind Singapore and Malaysia, 20 to 25 years behind South Korea and Taiwan, 50 to 75 years behind Japan in the field of Physics. What I mean is that it would take those many years of massive sustained development of science, manpower research and training facilities and research capabilities

in Physics in the Philippines before we could catch up with those countries' state of development in physics.

In his keynote address, President Angara gave a survey of the incredibly backward state of physics in our country that makes for the frustrating situation with which we are all familiar. What really makes it so frustrating is that we do not seem to have progressed at all for the past 20 years in our efforts to develop physics locally. The most glaring manifestation of the backward state of physics is the fact that up to now, physics research in our country is embarrassingly negligible whether in terms of quality or quantity.

Outside of the Philippine Atomic Energy Commission (PAEC) and the Manila Observatory, there is no other research laboratory in the Philippines and there is hardly any sustained research program elsewhere. In fact, aside from the on-going theoretical research activities at UP Physics Department there is no research work at the post-doctoral level that I know of which is going on in any other university and even the highly individualized research activities in the field of physics that have been going on at UP for the past 14 years have not yet blossomed into a research tradition that one can be proud of.

The poor state of the field of research in our country today can be attributed to four inter-related factors:

- a. Manpower lack
- b. The lack of university research facilities for physics
- c. The relative isolation of local physicists from international community of physics research
- d. The government impeding of basic research in physics that was mentioned by President Angara

We have at present only 21 Physicists in the entire country out of which only about 8 to 9 are actively engaged in physics research and these 8 to 9 research active physicists are not even concentrated in one institution. They are distributed among UP and Ateneo, at PAEC and one or two others, with UP having the largest concentration because of its research active Ph.D. emphasis. Thus, we have only about two research active Ph.D. physicists for every 6 million of our population as contrasted with the figure of 800 to 1,000 research active Ph.D. physicists for 10 million population in the advanced countries. In short, we are extremely far, far below the critical mass needed to generate a self-sustaining research tradition of maintaining a genuine research community in the field of Physics.

The lack of research facilities for physics in our universities and colleges has also worked against the promotion of physics research. For as a major

research in physics there is at present no available adequate laboratory in any university and even nowhere in the entire country is there a single physics library with a complete up-to-date collection of physics books, journals and abstracts; neither do we have a powerful computer available for computation work in physics research nor is there a fully equipped machine shop or electronic shop for maintaining, repairing, or fabricating sophisticated science equipment in physics.

Situated as we are in the backwaters of international science, we are enveigled in our research efforts by isolation from the research centers in physics. Research physicists are fully aware that research advances through the so-called invisible family of international networks of research groups which collaborate extensively on common research interests, exchange of research data and visit one another to discuss research progress. Contact with such an invisible network through frequent exchange and personal meetings is therefore of vital importance to local physics, but government policy does not seem to realize this. And this brings me to the last and greatest factor which is impeding the development of physics research in this country: the government's impeding of physics research. The National Science policy of concentrating its forte on applied research in agriculture, industry and trade, to the neglect and even discouragement of basic science like physics is the basic cause of the retardation of physics in this country.

The fact that the NSDB has not improved the local state of physics since its establishment 24 years ago in 1958 is a proof of this. Physics has always been looked down upon by government politicians as irrelevant, as something not worth developing. I think this attitude towards Physics in our country stems basically from a defeatist mentality, an inferiority complex that hides itself under the posture of economic progress. To technocrats and politicians, physics and high technology are too expensive, too difficult, too time-consuming for our country. So advanced research in physics should be left to the advanced countries. They believe that we should simply patronize the hand products of physics in the advanced countries and that means we merely import the technology products of physics for our own. This is the kind of thinking that has kept our basic sciences backward and that has left us far behind the indigenous technological and industrial capabilities of even our Asian neighbors.

This is the unbelievable situation that we are faced with in trying to develop physics in this country. Even with these initial conditions, it does not seem realistic to talk at all about national research projects in physics. But at the same time these same conditions demand that we try to maximize our

extremely limited research resources for the most effective local development of our discipline. In other words, we must adopt a selective research strategy. This entails the definition of our directions in physics, and our means of arriving at a well-defined goal. Now, when we talk of physics often we talk of relevance and I think that the word relevance is the most pernicious word when applied to the basic sciences because all politicians when they speak of relevance always think in terms of immediate overnight results. And this is the kind of thinking which has hampered the development of the basic sciences especially in physics. Instead of relevance, I propose that we direct the development of natural science toward the attainment of a long-term goal in technological self-reliance. That means an indigenous activity to contribute to world science and to develop our own technologies based on our own researches on the local scene. But of course given the initial conditions that we have at present we cannot hope to achieve such a long-term goal within a few years or even in 20 years, but at the same time we need a tremendous effort; otherwise, we can never achieve any self-reliance at all. So what should be our short-term goal? Well, obviously, because of the shortages of research manpower the immediate goal for the next 5 to 10 years should be the development of our research manpower. Hence, our research activity must necessarily be linked to our graduate programs especially our masteral and doctoral programs.

Now, given also a limited research manpower, there is a need to concentrate our research activities on selected areas. Next school year we will be starting the Ph.D. consortium program in physics at the University of the Philippines, Ateneo de Manila and De La Salle University. It has been the consensus within the three Physics Department of this consortium that at least for the next 5 to 10 years we shall concentrate our research efforts on three areas: condensed matter physics, quantumelectronics, field and particles.

Condensed matter physics is the expanded area of solid state physics. Solid state physics has expanded to cover also liquids so that it has come to be known now as condensed matter physics.

This is the area that we would like to concentrate on because we feel that the equipment needed for research in this area is not too expensive as compared to the equipment for higher physics which is way beyond what we can afford. At the same time condensed matter physics is something that directly applies to semi-conductor applications, microelectronics, and finally to materials of science and technology. Quantumelectronics, the second field of concentration in the consortium program covers the physics of the laser. As you know, the laser has various applications not only in the other fields of

natural science but also in industry. At the same time, this is a technology that we can do easily and even apply to our own needs in this country; that's why we feel that laser should be an area that ought to be developed in the consortium. And then the third area or field covers elementary particle physics, based on mathematical physics. This is an area that may not have any immediate application like the other areas but it is something that appeals to some of our graduate students and at the same time it is an area that keeps a window of the frontier of physics. And so we feel that it ought to be one of the concentrations in a consortium program. Now, of course, I'd like to emphasize that the selection of the three broad areas of concentration does not mean the exclusion of other areas or disapproval of such proposals in other areas. The need of prioritization of selection of research areas simply means that we concentrate our efforts on certain subjects because, as you know, in physics we really cannot foretell where the next breakthrough will come from and so we cannot deny and we must not neglect that some people who would like to research in some field, to do his thing in that that field. Now, when I talk of these areas of concentration, experimental and theoretical work will be encouraged in both condensed matter physics and quantumelectronics but in the field of physics of particles and fields experimental work will not be possible unless we can explore the possibility of cooperative schemes with our asian neighbors, even with China, for the setting up of regional accelerators and common facilities.

Now that only covers pure physics but, given the kind of thinking that we have in this country about relevance and so on, I think that we in the physics community should also look into applied physics research. In fact, in the UP physics department, we began our own Applied Science Program about a year ago. This is not yet a well-defined program but I'd like to propose that we should consider the following areas for applied physics. The applications of solar energy or solar scientific instrumentation, I believe, are areas where we in the physics community should start trying to fabricate some of the instruments that we still import from abroad. This is a matter of self-reliance. We should look also into the various possibilities for the application of microelectronics, and the usual microbiologies for industrial programs I have mentioned earlier. Now, what I firmly suggest is that the National Physics community should consider the three areas of concentration in the consortium and we would then concentrate on the national efforts in these areas as well as in the other suggested areas relating to applied physics.

As I talk about direction and concentration, I'd like to deal now with the need for collaboration in the development of physics research in our country

and that means the encouragement of collaboration among various researchers. The development of research groups, in other words, instead of doing research on an individualized basis. Let's start talking together exchanging ideas about possible research problems and in time form research groups that will attract common areas of research interest. Now, earlier, I touched on the problem of isolation and I think that this is very crucial in the field of physics research. We should strive to encourage more communication — both internal communication among ourselves and international communication with our neighbors and with the other research groups.

What I propose is that starting next summer we ought to start summer research seminars for specialized courses on a basis that will enable our graduate students and even some of our undergraduate students to participate. This is one way of encouraging research in our country and at the same time throughout the year we should have at least a monthly seminar by our research active physicists. In fact, in the consortium program there is a plan to hold a monthly seminar in order to acquaint our graduate students with possible research areas. To stimulate local interest in physics we should also try to get in professors and lecturers who can stay here for a few months or even longer to give special lectures on advanced topics and even start local programs for local physicists and graduate students.

Cooperation of our colleagues in the other asian countries should also be explored and worked out. In fact, the Samahang Pisika ng Pilipinas, a member of the Asian Institute of Physics, has started communicating with the officers of the asian-based organization in Kuala Lumpur for a directory of local research programs of activities in the different asian countries so that we will know for a start what our neighbors in this area are doing in physics; and if we find out that someone in Kuala Lumpur is doing research in which some of our people here are interested we can communicate or we could also start our own research program. In fact, I would also like to propose that starting next year when we hold our next National Physics Congress, we should change the character of our convention and conference. We should start presenting original papers in physics rather than traditional complaints about physics. Every year or so every time we meet we hear the same complaints, we hear the same gripes; I think, it is about time that we should start doing physics. We should start trying to improve the state of our research in physics. We should even start involving our students in some of our advanced programs. I believe that one of the best ways of attracting students into physics is to show students right from their first year in college that there are interesting activities going on in the physics department so that when we meet again

next year we will have more people and even students presenting papers on their research work in physics.

Now what are the immediate prospects for us in the physics profession in the next school year. Well, I have already mentioned the Ph.D. consortium to start in June of this school year with six Ph.D. fellows. Three of these fellows will be concentrating on quantumelectronics, while the other three will be specializing in solid state physics. I would also like to mention our plans for the UP Natural Science Center which are being favorably considered in National Science and Technology Authority and this is to expand the Natural Science Research Center into an NSDB-assisted National Research Center for the basic sciences including Physics. In fact, as already mentioned by President Angara in his keynote address, we are already setting up a laboratory for experimenting in solid state physics in the Natural Science Research Center. This laboratory is being set up by our newest Ph.D. at the UP Physics Department, Dr. L. Dy, a Ph.D. experimental solid state physicist from the University of Illinois. We hope to be able to get more Ph.D.'s in solid state physics for the next few years. In fact, we are sending many of our faculty members abroad to specialize in the field of solid state physics.

Now, what we have in mind for the National Science Research Center is that it is going to be a national center for basic scientists and physicists in other institutions outside UP who would like to do research. They can come to the center and do research in their respective fields, whether in physics, biology, chemistry, or Mathematics or even earth science.

During the summer we will also have research workshops and we will have a graduate faculty program where basic research scientists in other institutions can stay at the UP National Science Center for the entire summer, do anything they want, attend workshops, do research and get refreshed so that when they go back to their own institutions they can infect their students with the research fever. And so these are the initial steps being taken now within the consortium for renewed activities in basic research and so I hope that this congress will mark an upsurge in physics research so that by the time we meet again next year we can start talking about our research activities. We could start exchanging information on research ideas, or research projects rather than exchanging problems about our salaries, our textbooks, and so with that I would like to say that when we go back to our various institutions, we should start encouraging students to do physics. We should start trying to do research however small so that at least we can make a beginning toward a long-term goal of achieving self-reliance.

Thank you very much.